In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

In [2]:
```
url = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv

data = pd.read_csv(url)

data = data[['eruptions', 'waiting']]

print(data.head())
print(data.shape)
```

```
   eruptions  waiting
0      3.600       79
1      1.800       54
2      3.333       74
3      2.283       62
4      4.533       85
(272, 2)
```

In [5]:
```
X = data[['eruptions', 'waiting']].values
```

In [7]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

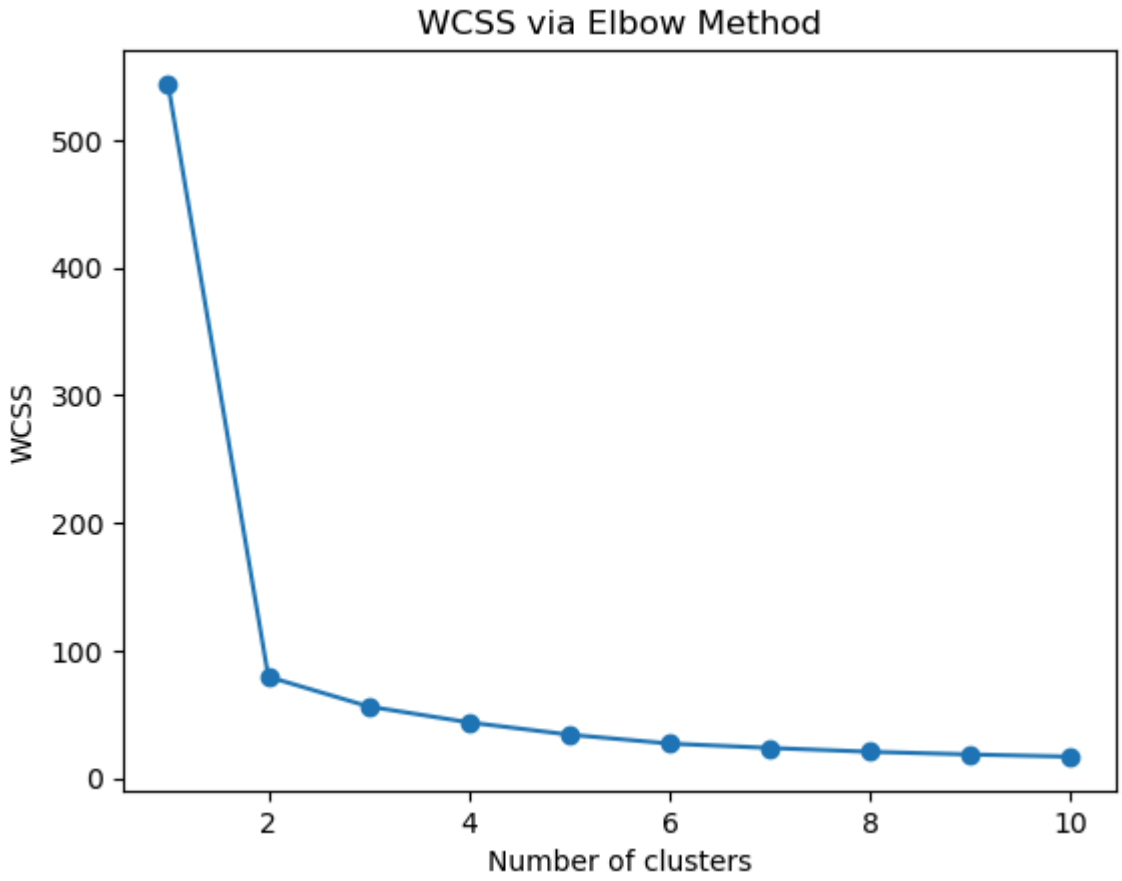
In [9]:
```
wcss = []

for i in range(1, 11):
    model = KMeans(
        n_clusters=i,
        init='k-means++',
        random_state=42,
        n_init=10
    )
    model.fit(X_scaled)
    wcss.append(model.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title('WCSS via Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```

```
C:\Users\Purnishamahi\anaconda3\Lib\site-packages\joblib\externals\loky\backend\c
ontext.py:136: UserWarning: Could not find the number of physical cores for the f
ollowing reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by se
tting LOKY_MAX_CPU_COUNT to the number of cores you want to use.
  warnings.warn(
  File "C:\Users\Purnishamahi\anaconda3\Lib\site-packages\joblib\externals\loky\b
ackend\context.py", line 257, in _count_physical_cores
    cpu_info = subprocess.run(
               ^^^^^^^^^^^^^^^
  File "C:\Users\Purnishamahi\anaconda3\Lib\subprocess.py", line 548, in run
    with Popen(*popenargs, **kwargs) as process:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Purnishamahi\anaconda3\Lib\subprocess.py", line 1026, in __init_
_
    self._execute_child(args, executable, preexec_fn, close_fds,
  File "C:\Users\Purnishamahi\anaconda3\Lib\subprocess.py", line 1538, in _execut
e_child
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\Purnishamahi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:142
9: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when t
here are less chunks than available threads. You can avoid it by setting the envi
ronment variable OMP_NUM_THREADS=2.
  warnings.warn(
C:\Users\Purnishamahi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:142
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here are less chunks than available threads. You can avoid it by setting the envi
ronment variable OMP_NUM_THREADS=2.
  warnings.warn(
```



In [11]:
```
model = KMeans(
    n_clusters=2,
    init='k-means++',
    random_state=42,
    n_init=10
)

y_kmeans = model.fit_predict(X_scaled)

print(y_kmeans)
```

```
[0 1 0 1 0 1 0 0 1 0 1 0 0 1 0 1 1 0 1 0 1 1 0 0 0 0 1 0 0 0 0 0 0 0 0 1 1
 0 1 0 0 1 0 1 0 0 0 1 0 1 0 0 1 0 1 0 0 1 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 0
 1 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 1 0 1 0 1 0 0 0 1 0 1 0 1 0 0 1 0 1 0 0 0
 1 0 0 1 0 1 0 1 0 1 0 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 1 0 0 0 1 0 1
 0 1 0 0 1 0 0 0 0 0 1 0 1 0 1 0 0 0 1 0 1 0 1 1 0 0 0 0 0 1 0 0 1 0 0 0 1
 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 1 0 0 1 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
 1 0 0 0 0 0 0 0 0 0 1 0 1 0 1 1 0 0 1 0 1 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1
 0 0 0 1 0 1 1 0 0 1 0 1 0]
```

```
C:\Users\Purnishamahi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1429: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
```

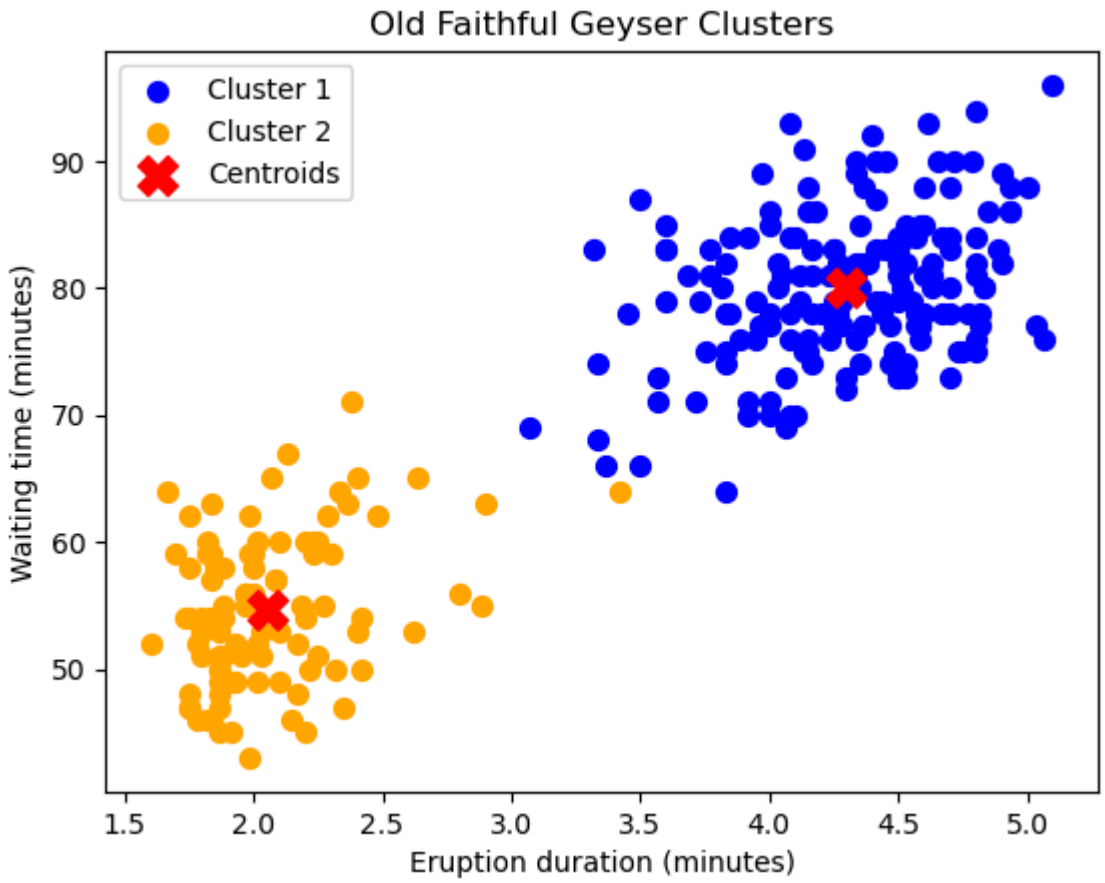
In [13]:
```
plt.scatter(
    X[y_kmeans == 0, 0],
    X[y_kmeans == 0, 1],
    s=50,
    c='blue',
    label='Cluster 1'
)

plt.scatter(
    X[y_kmeans == 1, 0],
    X[y_kmeans == 1, 1],
    s=50,
    c='orange',
    label='Cluster 2'
)

centers = scaler.inverse_transform(model.cluster_centers_)

plt.scatter(
    centers[:, 0],
    centers[:, 1],
    s=200,
    c='red',
    marker='X',
    label='Centroids'
)

plt.title('Old Faithful Geyser Clusters')
plt.xlabel('Eruption duration (minutes)')
plt.ylabel('Waiting time (minutes)')
plt.legend()
plt.show()
```


Old Faithful Geyser Clusters
Cluster 1
Cluster 2
Centroids
Waiting time (minutes)
90
80
70
60
50
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Eruption duration (minutes)

In [ ]: